atrocious. HF returns from 1980 to 2008 were 6.0% compared to 10.9% returns for the S&P 500 over the same time period. The 6.0% HF return barely beats the T-bill return of 5.6%. Not surprisingly, Dichev and Yu title their paper "Higher Risk, Lower Returns."

Some studies find positive outperformance for HFs of around 3% to 5% compared to standard market indexes.[29] (Some of these numbers are even statistically significant, especially for the top HFs.) A notable feature of HFs is that they exhibit *decreasing returns to scale* (just like mutual funds and private equity, which are covered in chapters 16 and 18, respectively).[30] This fact means that as HFs grow, they tend to have lower returns. Many of the outperforming funds are concentrated in the early years of the HF industry when they were small (see Figure 17.3). Zhong (2008) shows that the drop in HF alphas, especially in the post-2000 period, is the result of fewer funds capable of producing large, positive alphas as the industry has grown.

When the average return is computed for the whole HF industry, as Dichev and Yu (2011) do, large funds are given larger weights commensurate with their larger assets. When these large funds underperform, the losses to the overall HF industry are much larger. The aggregate losses of the HF industry led Simon Lack (2012), an ex-hedge fund manager, to begin his book with the taunting sentence: "If all the money that's ever been invested in hedge funds had been put in treasury bills instead, the results would have been twice as good."

Lack shows that in 2008, the HF industry lost investors more money than it made them over the previous decade.

Many investors are drawn to HFs by the record of returns (sketchy as it is), which was produced largely when the industry was nascent and many HFs were small. Risks in the early years were high, but early investors prospered—they obtained high rewards ex post for bearing very large ex-ante risk. The HF industry has since matured, and true outperformers are harder to find. Often the best HFs are small. But many asset owners are reluctant to select small, unproven HFs. They gravitate instead toward large HFs with long track records, vast assets under management, high-quality infrastructure, and better reporting and risk management. Of course the returns on these large funds aren't as high, on average, as small ones: they're less risky.

Persistence

Knowing that some HFs do well is useless unless we can predict which HFs do well. Academics are divided on whether persistence exists in HF returns. Malkiel and Saha (2005) report that the fraction of the top half of HFs that end up in

[29] See, among others, Brown, Goetzmann and Ibbotson (1999), Ibbotson and Chen (2006), Kosowski, Naik, and Teo (2007), and Jagannathan, Malakhov, and Novikov (2010).

[30] See Agarwal, Daniel, and Naik (2009), Boyson (2008), Zhong (2008), Fung et al. (2008), Dichev and Yu (2011), Getmansky (2012), Mozes and Orchard (2012), and Ramadorai (2013).